Triplet Contrastive Learning Framework With Adversarial Hard-Negative Sample Generation for Multimodal Remote Sensing Images

Zhengyi Chen, Chunmin Zhang, Biyun Zhang, and Yifan He

***Abstract*— Supervised learning models have achieved remarkable success in the field of remote sensing, but their applicability is limited by the significant requirement of high-quality labeled data. This article presents the triplet adversarial contrastive learning (TACL) model, as a self-supervised feature extractor. Considering the potential contrastive semantic conflicts, which may occur due to the different descriptive abilities of various modalities, TACL constructs a triplet contrastive learning framework aligning the two original modalities and a fused modality. To augment the acquired representations of the model, TACL introduces an adversarial hard-negative sample generation (AHSG) strategy, aiming to boost the resemblance between the feature vectors of negative samples and anchors. In addition, a ConvNeXt-based lightweight encoder is designed as the foundational backbone of the model, specifically enriching the expression of central features. A series of few-shot classification experiments substantiate the exceptional performance of the features extracted by TACL, with the simplistic classifier support vector machine (SVM). As a label-free pretraining approach, TACL holds great potential for enhancing the performance of various multimodal remote sensing tasks in scenarios with limited label availability.**

***Index Terms*— Contrastive learning, few-shot classification, multimodal remote sensing images, self-supervised learning.**

I. Introduction

THE emergence of remote sensing platforms provides a wealth of image information for Earth observation in a variety of modalities, accompanied by applications in many fields, such as environmental monitoring. Each modality of remote sensing images has its distinct advantages. hyperspectral images (HSIs), for instance, are ideal for areas requiring precise determination of substance components [1], [2]. LiDAR images can provide highly accurate and 3-D information on the Earth's surface structure and topography [3], and polarimetric images offer detailed information on the surface roughness and texture of targets [4].

Combining multimodal remote sensing images to process various downstream tasks has become a trend in recent years [5]. For example, Sun et al. [6] fused the very-high-resolution (VHR) images and LiDAR images for semantic segmentation by a multimodal fusion network. Ma et al. [7] and Xiong et al. [8] separately devised image fusion networks for target detection, leveraging thermal features from infrared images along with RGB images, and incorporating depth images along with RGB images. Zhang et al. [9] further extended it to the fusion of RGB and other arbitrary modality. Luppino et al. [10] proposed a deep image translation network for multimodal change detection. Li et al. [11] developed a multiscale attention mechanism-based ConvLSTM network associating multisource remote sensing information for land cover classification. Some semi-supervised networks represented by the X-ModalNet proposed by Hong et al. [12] have also achieved good results multimodal remote sensing tasks. However, in remote sensing applications, particularly in multimodal remote sensing scenarios, acquiring a sufficient amount of labeled data is expensive and time-consuming, which presents challenges for utilizing supervised learning-based models.

Self-supervised learning, serving as a label-free pretraining paradigm, has made remarkable strides in the field of computer vision. It unveils the intrinsic data characteristics via specific proxy tasks, with masked learning and contrastive learning standing out as the most extensively employed approaches. Masked learning involves the selective obscuring or "masking" of certain input features or components during the training process to enhance model generalization and robustness [13]. Contrastive learning aims to leverage the internal correlations within data to the maximum extent possible in order to attain more discriminative representations [14], [15], [16], [17], [18]. Recently, the concept of self-supervised pretraining has also been successfully implemented in multimodal fusion and representation learning of remote sensing images. Yang et al. [19] combined masked learning with contrastive learning and proposed the contrastive masked autoencoder (CoMAE) for the pretraining of small-scale RGB-D images. Chen and Bruzzone [20] devised a contrastive learning approach for image-level and super-pixel SAR-optical fusion. For remote sensing image semantic segmentation tasks, Li et al. [21] proposed the global style and local matching contrastive

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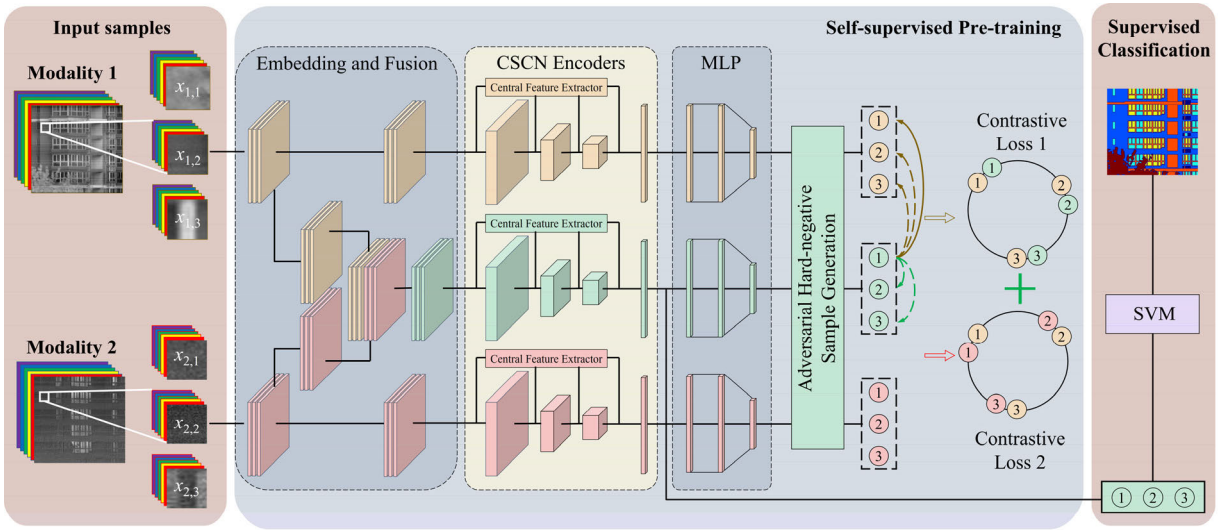


Fig. 1. Architecture of the proposed triplet adversarial contrastive learning (TACL) model.

learning network (GLCNet), which integrated both global and local features with contrastive learning modules to enhance image-level and local region representation. Meanwhile, Zhu et al. [22] applied contrastive learning on HSI classification tasks and validated its potential in few-shot classification. In general, contrastive learning generates two different views of input data by applying various data augmentations for comparison. However, in the case of multimodal data, they inherently possess two or more views, making them naturally suitable for contrastive learning frameworks. Leveraging this advantage, Xue et al. [23] established the self-supervised feature learning (S2FL) architecture with contrastive learning framework, extracting high-level feature representations from multimodal remote sensing data. Hang et al. [24] proposed the cross-modality contrastive learning (CMCL) model with a dual fine-tuning strategy for classification on HSI and LiDAR images with small numbers of training samples. Based on similar concepts, Wu et al. [25] and Jia et al. [26] developed the dual-encoder contrastive learning (DECL) model and the collaborative contrastive learning (CCL) model, respectively.

However, these methods have not taken into consideration the divergent scene descriptive capabilities of different modalities, which may result in semantic conflicts during the construction of contrastive learning. Furthermore, the quality of negative samples is also overlooked, as both excessively similar and easily distinguishable negative samples will undermine the effectiveness of contrastive learning. In this article, a self-supervised feature extractor called triplet adversarial contrastive learning (TACL) is proposed. TACL considers the two modalities as two distinct views of the target scene and merge them through a view fusion module. Subsequently, contrastive learning is applied between each individual view and the fused representation to address the potential contrastive semantic conflicts. TACL also employs an adversarial hard-negative sample generation (AHSG) strategy, to fully utilize negative samples that are too easily distinguishable from anchors. The architecture of the proposed TACL model is shown in Fig. 1. The main contributions of this article are summarized as follows.

1) The triplet contrastive learning framework for multimodal remote sensing images is proposed aiming at addressing the contrastive semantic conflicts. Within this framework, a view fusion module as well as the cross-view attention block (CVAB) is designed.
2) A simple and readily deployable AHSG strategy is developed to enhance the similarity between easily separable negative samples and anchor samples.
3) A central-enriched slight ConvNeXt (CSCN) encoder based on the ConvNeXt network is designed, incorporating a central feature extraction module (CFEM) that emphasizes center pixels in each input patch.

The rest of this article is structured as follows. Section II reviews contrastive learning briefly and introduces the motivations of our work. Section III presents the proposed TACL model in detail. Section IV gives the experimental results. Section V concludes this article.

II. Preliminaries and Motivations

A. Contrastive Learning

Contrastive learning, as a technique that aims to learn data representations by contrasting different views without requiring labeled data, has been successfully applied in several fields, including computer vision and natural language processing. Ye et al. [14] proposed the invariant and spreading instance (InvaSpread) model training to maximize the similarity between the anchor and positive samples while minimizing the similarity between the anchor and a set of negative samples. The simple framework for contrastive learning of visual representations (SimCLR) model proposed by Chen et al. [15] used a wider variety of data augmentations and a larger batch size to construct the contrastive views and added a multilayer perceptron (MLP) layer after the encoder. To ensure consistency of negative features in each iteration, the momentum contrast (MOCO) model proposed by He et al. [16] adopted the momentum update strategy to

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gradually progress the parameters of the encoders, updating one branch normally while applying momentum to the other. Grill et al. developed the bootstrap your own latent (BYOL) model that achieved the avoidance of negative samples through the mutual prediction of two augmented views [17]. On the basis of these works, He et al. [18] further proposed the simple Siamese representation (SimSiam) model, which employs a twin network structure and a stop-gradient operation on one of its routes, achieving good performance without requiring large batch sizes or momentum encoders.

Contrastive learning usually involves training a model to differentiate between similar and dissimilar pairs of data, without relying on explicit labels or supervision. Current mainstream contrastive learning models incorporate multiple data augmentation techniques to create multiple views of each sample. Let $x_i \in \mathbb{R}^{H\times W\times D}$ denote the ith sample of dataset. Two random transformations $t_1(\cdot)$ and $t_2(\cdot)$ will be applied on x_i to generate the augmented views $x_{1,i} = t_1(x_{1.i})$ and $x_{2,i} = t_2(x_{2.i})$. The transformations $t_1(\cdot)$ and $t_2(\cdot)$ are compilations of several augmentation methods, such as random cropping, random rotating, random flipping, and random noise.

These views are then encoded by the encoders $f_1(\cdot)$ and $f_2(\cdot)$, and views created by augmenting the same sample are treated as positive pairs in the latent space, while other samples are treated as negative samples. The SimCLR model found that adding a multilayer perceptron (MLP) as the projection head $g(\cdot)$ after the encoder can significantly improve the performance. Therefore, the feature vectors of two views in the latent space are $u_i = g_1\big(f_1\big(x_{1,i}\big)\big)$ and $v_i = g_2\big(f_2\big(x_{2,i}\big)\big)$.

In order to minimize the distance between anchors and their corresponding positive samples in the latent space, and to maximize the distance between anchors and their corresponding negative samples, the InfoNCE loss is typically employed in contrastive learning, which is defined as [15]

$$L_u(i) = -\log \frac{\exp(\mathrm{sim}(u_i, v_i)/\tau)}{\exp(\mathrm{sim}(u_i, v_i)/\tau) + \sum_{u_j^- \in \Lambda^-} \exp\big(\mathrm{sim}\big(u_i, u_j^-\big)/\tau\big)} \tag{1}$$

where u_i denotes the ith anchor of the first view, v_i denotes the corresponding positive sample of u_i, Λ^- denotes the set of negative samples of u_i, $\mathrm{sim}(\cdot,\cdot)$ denotes the cosine similarity of two feature vectors, and τ denotes a temperature parameter. The total loss of a whole minibatch B can be written as

$$L = \sum_{i\in B} [L_u(i) + L_v(i)]/2. \tag{2}$$

By leveraging contrastive learning frameworks, the pretrained networks can be utilized as excellent feature extractors. The features extracted by these networks have been proven to be highly effective for various downstream tasks, such as classification, semantic segmentation, and object detection.

B. Motivations

In pixel-level remote sensing tasks, a small patch centered on the target pixel is commonly used as an anchor, while patches centered on other pixels are treated as negative samples. However, there is often an excess of redundant information in the spatial dimension, with a significant number of pixels that are similar or even identical. In addition, in the case of multimodal data, different modalities may have varying levels of distinguishability for the same scene, and utilizing a contrastive learning framework directly could result in semantic conflicts.

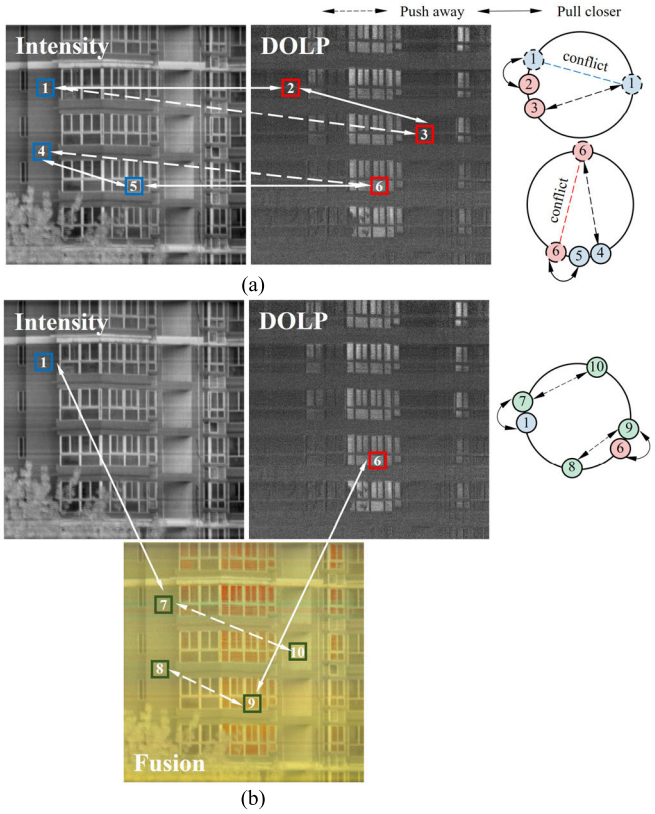


Fig. 2. Illustration of the contrastive semantic conflicts. The schematics on the far right indicate the spherical latent space where each patch is located after being encoded by the network. Semantic conflict occurs on Patches 1 and 6, but no longer exists with the fused view. (a) Contrastive learning between two original modalities. (b) Contrastive learning between the fused modality and two original modalities.

In this article, the contrastive semantic conflicts refer to the inconsistencies or the disparities that emerge within a latent space following encoding through contrastive learning, between the intended similarities and dissimilarities of representations. These conflicts reflect the difficulty in achieving the desired relative positioning of samples in the latent space, particularly when considering multiple objects and their corresponding modalities.

Fig. 2 gives an example of contrastive semantic conflicts. It illustrates the intensity image and the degree of linear polarization (DOLP) image of the same scene obtained by an imaging spectropolarimeter [27]. The term "pull closer" in Fig. 2 refers to the concept that, following encoding through contrastive learning, two patches can become similar in the new spherical latent space. As shown in Fig. 2(a), Patches 1 and 2 correspond to representations of the same object under different modalities. Guided by the principles of contrastive learning, the objective is to make them very close to each other within the latent space. Patches 1 and 3 correspond to

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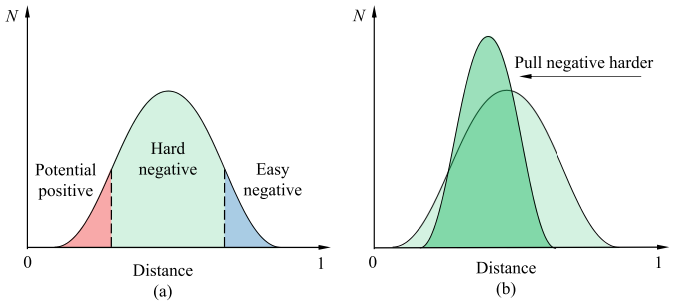


Fig. 3. Illustration of the adversarial hard-negative sample generation process. (a) Original distribution of the normalized Euclidean distances between negative samples and an anchor sample. (b) Distribution of the generated new negative samples.

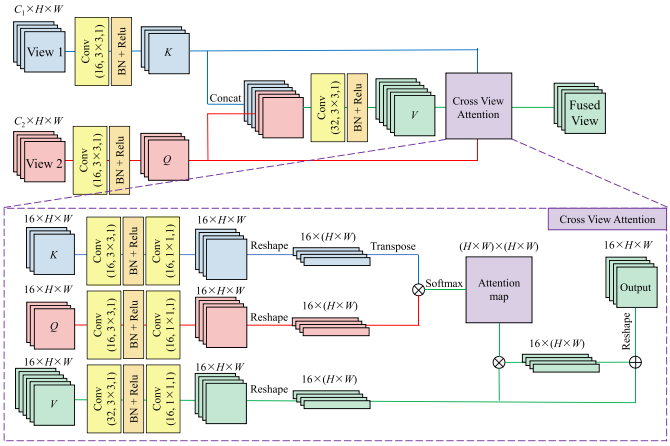


Fig. 4. Histograms of the view fusion module and the cross-view attention block (CVAB).

different objects, and their representations should be dissimilar within the latent space. However, Patches 2 and 3 exhibit minimal distinction in the original DOLP representation, and their encoded forms almost overlap within the latent space. For Patch 1, it is difficult to simultaneously pull it closer to Patch 2 while also pushing away from Patch 3. This conflict also occurs in Patch 6, which should be pulled closer to Patch 5 and, meanwhile, be pushed away from Patch 4. However, Patches 4 and 5 exhibit nearly identical characteristics in the intensity view, naturally exhibiting proximity in the latent space.

To prevent semantic conflicts arising from differences in the representation of different modalities, this article attempts to use a fused modality to construct contrastive learning separately with each of the initial modalities, as shown in Fig. 2(b). After fusion, Patches 7 and 10 (corresponding to Patches 2 and 3 in DOLP) exhibit notable distinctions, leading to their distinct positions within the latent space without overlap. Patch 1 is capable of simultaneously being pulled closer to Patch 7 while being pushed away from Patch 10. Therefore, the semantic conflicts are no longer present with the fused view. In one modality, certain pixels corresponding to different targets may not be distinguishable, whereas in another modality, they may be discernible and vice versa. The fused modality has the potential to distinguish all types of targets, within the range that can be distinguished from each of the two original views of course. Based on this concept, we proposed the triple contrastive learning framework, which will be described in Section III-A.

One of the primary goals of traditional contrastive learning is to enlarge the distance between negative samples and anchors. However, in real-world data distribution, there exist numerous "easy-negative" samples that are already distinct from anchors, yet these particular negative samples remain underutilized. Here, we divide negative samples into easy-negative and hard-negative based on their normalized Euclidean distances to the anchor, as shown in Fig. 3(a). The more similar a sample is to the anchor, the harder it is to be distinguished. Some researchers have found that choosing negative samples that are more similar to the anchor can lead to an improvement in the effectiveness of pretraining networks [28], [29]. On the other hand, for samples that are too similar to the anchor, due to the spatial redundancy of remote sensing images, they are likely to be the same type of land cover as the anchor itself, i.e., "potential-positive" samples. Obviously, it is inappropriate to retain these potential positive samples in the set of negative samples.

To make full use of these "easy-negative" samples and to mitigate the interference from "potential-positive" samples, this article introduces a strategy for updating the features of these negative samples. The primary role of this strategy is to transform the encoded features with the aim of making the easy-negative samples "harder," as shown in Fig. 3(b). This strategy goes against the training objective of contrastive learning, and therefore, we refer to it as AHSG.

III. Methodology

In this section, we will introduce the proposed TACL model in detail, which consists of three parts: the triplet framework for contrastive learning, the AHSG strategy, and design of network structure. As a self-supervised method, this model's training process does not require labels. Pixels of remote sensing images will be transformed into feature vectors by this pretraining network, and the goal is to increase the discriminability of these feature vectors so that they can be more effectively applied to various pixel-level downstream tasks such as land cover classification.

A. Triplet Framework for Contrastive Learning

As shown in Fig. 1, for each individual pixel, a patch of size $H \times W$ is taken with the pixel at its center. These patches are then used as samples and inputted into the network sequentially. For the convenience and intuitiveness of network design, all input patches are resized to 16×16. Patches from the two original modalities will first be passed through their respective embedding blocks, which consist of a 3×3 convolutional layer, a batch normalization layer, and the ReLU activation layer.

To avoid semantic conflicts, as described in Section II, a fused view is required, which will be generated by the designed view fusion module. The view fusion module is constructed in Fig. 4. The parameters of the illustration of convolutional layers are (number of output channels, kernel

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size, stride). After being convolved again, the two embedded views are concatenated along the channel dimension and undergo initial fusion through another convolution layer. These views are further integrated through a designed cross-view attention block (CVAB). CVAB is developed from the self-attention mechanism, which has been widely applied in recent years. The traditional self-attention mechanism multiplies the input vector I by three sets of weight coefficients to obtain the query Q, the key K, and the value V, i.e., $Q = W_q I$, $K = W_k I$, and $V = W_v I$ [30], [31]. The final output is defined as

$$A = V\,\mathrm{soft\,max}\left(\frac{K^T Q}{\sqrt{D_k}}\right) \tag{3}$$

where D_k is the dimension of the input vector. The traditional self-attention mechanism applies different transformations to the same input as Q, K, and V. The proposed cross-view attention treats two source views as query Q and key K to compute an attention map, which is then used as spatial weights assigned to each pixel of the fused view.

The generated fused view and the two source views will be encoded by their respective backbone networks. This article proposes an encoder based on the ConvNeXt structure [32], [33], which will be described in detail in Section III-C. Similar to SimCLR and SimSiam, an MLP layer is added to the back of the encoder.

In the TACL framework, the fused view will be compared to the two source views separately in the latent space, to define self-supervised loss functions for training network parameters. The normalized feature vectors corresponding to patches at the same spatial position in two views are defined as a positive sample pair, while the feature vectors of other spatial positions are their negative samples. Let u_i, v_i, and f_i denote the feature vectors of the ith sample of the two source views and the fused view in the latent space, respectively. The loss functions, based on the InfoNCE loss, between the fused view and one of the source views are modified as

$$L_{u,f}(i) = -\log \frac{\exp(\mathrm{sim}(f_i, u_i)/\tau)}{\sum_{u_j^- \in \Lambda_u^-} \exp\left(\mathrm{sim}(f_i, u_j^-)/\tau\right)} \tag{4}$$

$$L_{v,f}(i) = -\log \frac{\exp(\mathrm{sim}(f_i, v_i)/\tau)}{\sum_{v_j^- \in \Lambda_v^-} \exp\left(\mathrm{sim}(f_i, v_j^-)/\tau\right)} \tag{5}$$

where Λ_u^- and Λ_v^- denote the negative sample sets. Then, the total triplet contrastive loss of a minibatch B is

$$L = \frac{1}{N}\sum_{i \in B}\left(\frac{1}{2}l_{u,f}(i) + \frac{1}{2}l_{v,f}(i)\right) \tag{6}$$

where N is the batch size.

The proposed triplet framework does not require augmentations on the patches of the source view to obtain two views for comparison, but data augmentation, as a widely adopted technique to improve the robustness of networks, is still retained in this method. To be specific, the two source views will still undergo data augmentations consisting of random cropping, random rotating, random flipping, and random noise before being input to the embedding layers.

B. Adversarial Hard-Negative Sample Generation (AHSG) Strategy

In principle, having more negative samples generally leads to improved representational ability. However, too many negative samples can result in excessive computational costs. SimCLR utilizes only the data within the same minibatch as negative samples [15], whereas MOCO stores the feature vectors of all negative samples in a memory bank, randomly selecting a small subset for each computation and updating the memory bank with momentum encoding [16]. These methods did not take into account the spatial redundancy of remote sensing images, as well as the quality of negative samples, specifically referring to their similarity to anchor samples. As mentioned in Section II, this article proposed an adversarial update method to generate negative samples that are more difficult to distinguish from the anchor.

The first step is the initial selection of negative samples. Considering the nonlocal redundancy of remote sensing images, we cluster the pixels into n_c clusters using k-means with the HSI view and then select the samples closest to the cluster centers as the initial negative samples. In each epoch, these negative samples will be fed to the pretraining network without gradient backpropagation. In the experiments, the number of clusters n_c is set as 300, implying that all batches share a common negative sample set with a size of 300.

In our method, the specific approach employed for the selection of initial negative samples is relatively unimportant, as long as it can ensure that the distribution of samples does not exhibit excessive extremeness. In fact, even directly conducting random pixel sampling from images or uniformly sampling across spatial dimensions is considered acceptable. Here, k-means is adopted to avoid the potential presence of an extremely long-tail distribution, where certain specific classes might be overly abundant, while others remain disproportionately scarce. Furthermore, choosing the HSI-based selection method rather than relying on the fused view is attributed to their near-identical performance, with the HSI-based approach being simpler.

Converting negative samples being harder refers to making the feature vectors of negative samples more similar to anchors. Taking the comparison between the first source view and the fusion view as an example, for each negative sample, the goal can be expressed as a minimization problem

$$\min_{\widehat{u}_j} \left\| f - 1\widehat{u}_j \right\|_2^2 \tag{7}$$

where $\widehat{u}_j$ denotes the generated new feature vector of the jth negative sample and 1 denotes the column vector with all elements being one.

With (7), all negative samples will collapse into the same feature vector, i.e., the mean vector of f, which violates the principle of contrastive learning. To avoid excessive disruption of the original data distribution, it is necessary to add a regularization term

$$\min_{\widehat{q}_j} \left\| f - 1\widehat{u}_j \right\|_2^2 + \lambda \left\| \widehat{u}_j - u_j \right\|_2^2 \tag{8}$$

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where λ is a regularization parameter, and the solution can be obtained by taking the derivative with respect $\widehat{u}_j$ being zero

$$\widehat{u}_j = \left(1^T 1 + \lambda\right)^{-1}\left(1^T f + \lambda u_j\right). \tag{9}$$

It is evident that this solution is a weighted sum of f_{mean} (the mean vector of f) and the initial u_j, with the weighting balanced by the regularization parameter λ. f_{mean} is expected to represent the average features of various types of land cover targets. However, remote sensing images are spatially redundant, and the feature vectors of the same class of land cover targets are usually similar. In other words, f_{mean} will be more influenced by the classes that appear more frequently, which can lead to poor performance when the sample distribution is uneven. Considering this, based on the graph theory, we construct the adjacency matrix of f by $\text{AM} = ff^T$ and the degree matrix by $\text{DM} = \text{diag}(\text{sum}(\text{AM}, \text{dim} = 1))$. The degree matrix here represents the sum of cosine similarities between each vector and all other vectors in f. If the value of $\text{DM}(i, i)$ is larger, the ith feature vector in f should be assigned a smaller weight. Then, the target function (8) becomes

$$\min_{\widehat{q}_j} \left\| W\left(f - 1\widehat{u}_j\right)\right\|_2^2 + \lambda \left\|\widehat{u}_j - u_j\right\|_2^2 \tag{10}$$

where $W = (\text{DM}/\max(\text{DM}))^{-1}$. Also, the solution of (10) is

$$\widehat{u}_j = \left(1^T W^T W 1 + \lambda\right)^{-1}\left(1^T W^T W f + \lambda u_j\right). \tag{11}$$

Similarly, for the second source view, we have

$$\widehat{v}_j = \left(1^T W^T W 1 + \lambda\right)^{-1}\left(1^T W^T W f + \lambda v_j\right). \tag{12}$$

A problem that cannot be ignored is the value of λ, which determines the distributions of $\widehat{u}$ and $\widehat{v}$. Observing (10), the solution can be also written as

$$\begin{aligned}\widehat{u}_j &= \sum_{i\in B} \frac{W(i,i)^2}{\sum_{i\in B} W(i,i)^2 + \lambda} f_i + \frac{\lambda}{\sum_{i\in B} W(i,i)^2 + \lambda} u_j \\ &= \sum_{i\in B}^{c_i f_i +} c_0 u_j\end{aligned} \tag{13}$$

where $c_0 + \sum_{i\in B} c_i = 1$. The solution can also be regarded as a weighted sum of f and u_j, with the weighting balanced by c_0 and $\cup_{i\in B} c_i$. Obviously, the larger the value of c_0, the closer $\widehat{u}_j$ is to the initial u_j. Here, we determine the optimal value of λ through c_0 to reduce the influence of different data distributions and different experimental settings:

$$\lambda = \frac{c_0}{1 - c_0} \sum_{i\in B} W(i,i)^2. \tag{14}$$

To visualize the impact of the AHSG process on negative samples, one method is to randomly generate some anchors and negative samples and compare the normalized Euclidean distances between the anchors and negative samples before and after the AHSG process. To better match the distribution of real-world data, 512 pixels and 300 pixels of the Houston 2013 dataset are randomly selected, as examples of anchors and negative samples, respectively. Also, their hyperspectral data are regarded as the corresponding feature vectors. Fig. 5 shows the statistical histogram of the normalized Euclidean distance between all generated new negative samples and a certain anchor.

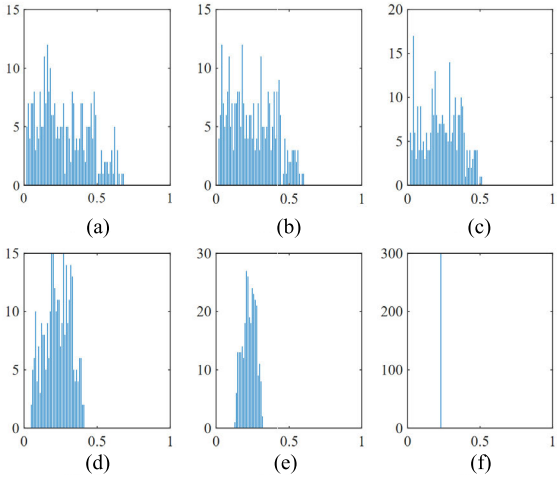


Fig. 5. Histograms of normalized Euclidean distances between the generated new negative samples and one of the anchors under varying values of the parameter c_0. (a) Original. (b) $c_0 = 0.8$. (c) $c_0 = 0.6$. (d) $c_0 = 0.4$. (e) $c_0 = 0.2$. (f) $c_0 = 0$.

The updating of negative samples, as can be seen in Fig. 5, achieves two objectives: 1) negative samples that were far away from the anchor are pulled closer, making them "harder," and 2) negative samples that were too close to the anchor are pushed further away, preventing them from being potential-positive. As c_0 decreases, the data distribution becomes increasingly concentrated and ultimately converges. In the following experiments, c_0 is set as 0.95, and the influence of the value of c_0 on the final experimental performance will be further examined in Section IV.

C. Design of the Backbone Encoder

This subsection provides a detailed description of the CSCN encoder of the proposed TACL model. The encoder is expected to be both lightweight and efficient, allowing for the extraction of spatial and spectral features from the target patches simultaneously.

The mainstream encoders used for visual modality can be roughly divided into two types: convolution-based encoders and transformer-based encoders. The primary advantage of transformer-based encoders over traditional convolutional methods is their ability to capture global spatial information, and thus, it is more suitable for large-scale images. However, this advantage is not significant for small patch sizes such as 16×16. In addition, ConvNeXt, as a pure convolutional model, achieves competitive performance in various visual tasks by using an inverted bottleneck module and larger kernel sizes [32], [33]. Therefore, the proposed model draws inspiration from the ConvNeXt architecture to construct the encoder. Since the input patch size is only 16×16, the kernel size for convolution remains at 3×3 instead of the 7×7 in the ConvNeXt block (CNB). For differentiation, here the version with retained small-size convolutional kernels is referred to as the slight CNB (SCNB).

For the three views, their encoders share the same network structure with different weight parameters. The detailed design of the encoders is shown in Fig. 6. The main brunch is a

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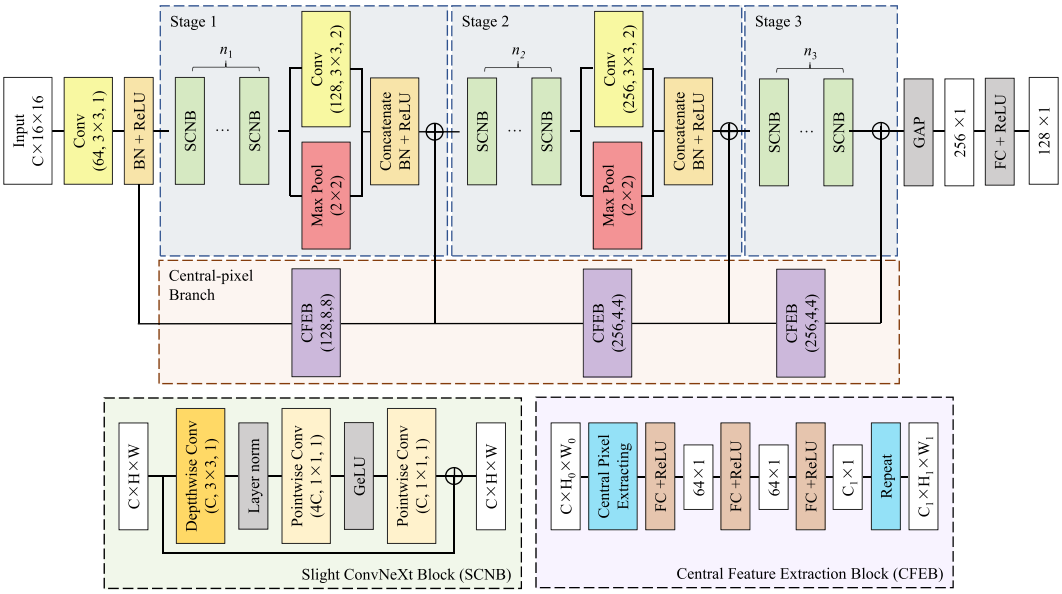


Fig. 6. Architecture of the central-enriched slight ConvNeXt (CSCN) encoder. The encoder is divided into three stages, each including several slight ConvNeXt blocks (SCNB) and a parallel central feature extraction block (CFEB). Downsampling is used between every two stages to halve the patch size and double the number of channels.

three-stage pyramid structure in which the spatial size will be halved and the number of channels will be doubled at each stage. There are multiple SCNB in each stage, with the number of blocks being $[n_1, n_2, n_3]$. Following the first and second stages, downsampling is achieved through a 2×2 max pooling layer and a 3×3 convolution layer with a stride of 2. As a result of two rounds of downsampling operations, the output size of the third stage will be $256 \times 4 \times 4$. Through a global average pooling (GAP) layer and a fully connected (FC) layer, the input patch is ultimately transformed into a 128×1 feature vector.

As mentioned before, a 16×16 patch surrounding each pixel is treated as the original input feature of the corresponding pixel. Obviously, the central pixel of this patch, i.e., the target pixel itself, contains the most crucial information. The other pixels within the same patch may not belong to the same land cover species as the central pixel, as shown in the patches $x_{1,3}$ and $x_{2,3}$ in Fig. 1. Therefore, a central pixel branch is designed to enhance the encoder's representation of the central pixel's information. The central pixel branch consists of three central feature extraction blocks (CFEBs), corresponding to the three stages mentioned above, as shown in Fig. 6. In each block, assuming that the input size is $C \times H_0 \times W_0$, only the central pixel, with a size of $C \times 1$, will be extracted. After passing through three FC layers, its channel number becomes C_1, and then, it is shifted and duplicated in two spatial dimensions to form a tensor with a size of $C_1 \times H_1 \times W_1$. The output sizes (C_1, H_1, W_1) of the three central pixel blocks are (128, 8, 8), (256, 4, 4), and (256, 4, 4), respectively. The output of each central pixel block will be added to the output of the corresponding stage, serving as the input to the next stage.

IV. Experiments

A. Datasets Description

In the experiments, three multimodal datasets are utilized to evaluate the performance of different methods. The corresponding labeled categories as well as the corresponding number of pixels in ground truth maps are shown in Table I.

1) *Houston 2013 HSI-LiDAR Dataset:* This dataset was distributed in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest. It was gathered a compact airborne spectrographic imager over the University of Houston campus and the surrounding rural areas with 1905×349 pixels. The HSIs consist of 144 spectral bands ranging from 364 to 1046 nm, and the LiDAR image covers the same area. The pseudo-color image and the LiDAR image, as well as the ground truth map, are shown in Fig. 7(a).
2) *Trento HSI-LiDAR Dataset:* This dataset was collected by the AISA Eagle sensor over a rural area in Trento, Italy. It contains 166×600 pixels with a spatial resolution of 1 m. The HSIs consist of 63 spectral bands ranging from 402.89 to 989.09 nm, and the LiDAR image also covers the same area. The pseudo-color image and the LiDAR image, as well as the ground truth map, are shown in Fig. 7(b).
3) *Augsburg HSI-SAR Dataset:* This dataset comprises HSI acquired by the HySpex sensor and synthetic aperture radar (SAR) data acquired by the Sentinel-1 sensor in Augsburg, Germany. The HSIs consist of 180 spectral bands ranging from 400 to 2500 nm, and the SAR image covers the same area with 332×485 pixels. The pseudo-color image and the SAR image, as well as the ground truth map, are shown in Fig. 7(c).

B. Experimental Settings

The HSIs will be preprocessed using the principal component analysis (PCA), and only the first ten principal components are retained. All experiments are carried out on a workstation with the Intel(R) Xeon(R) Gold 6226R CPU and the Nvidia GeForce RTX 3090 GPU. Furthermore, some other details of experimental settings are as follows.

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TABLE I

LAND COVER TYPE, THE NUMBER OF TRAINING SAMPLES, AND TESTING SAMPLES OF THREE MULTIMODEL REMOTE SENSING DATASETS

Houston 2013 HSI-LiDAR				Trento HSI-LiDAR			
No.	Class	Train	Test	No.	Class	Train	Test
1	Healthy grass	5	1246	1	Apples	5	4029
2	Stressed grass	5	1249	2	Buildings	5	2898
3	Synthetic Grass	5	692	3	Ground	5	474
4	Tree	5	1239	4	Woods	5	9118
5	Soil	5	1237	5	Vineyard	5	10496
6	Water	5	320	6	Roads	5	3169
7	Residential	5	1263		Total	30	30184
8	Commercial	5	1239	**Augsburg HSI-SAR**			
9	Road	5	1247	1	Forest	5	13502
10	Highway	5	1222	2	Residential Area	5	30324
11	Railway	5	1230	3	Industrial Area	5	3846
12	Parking Lot 1	5	1228	4	Low Plants	5	26852
13	Parking Lot 2	5	464	5	Allotment	5	570
14	Tennis Court	5	423	6	Commercial Area	5	1640
15	Running Track	5	655	7	Water	5	1525
	Total	75	14954		Total	35	78259

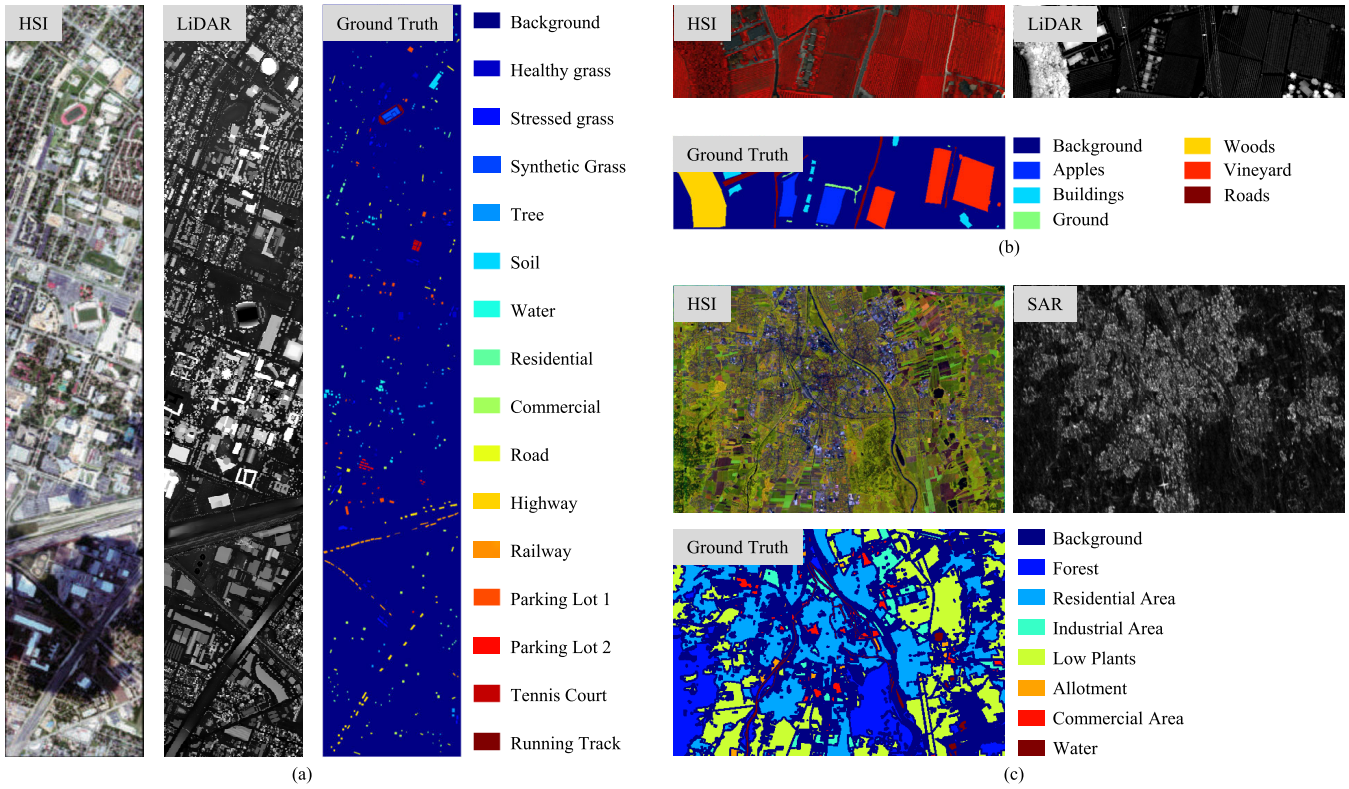


Fig. 7. Three multimodel remote sensing datasets. (a) Houston 2013 HSI-LiDAR dataset. (b) Trento HSI-LiDAR dataset. (c) Augsburg HSI-SAR dataset.

Pretraining With Superpixels: Due to the large scale and redundancy of optical remote sensing images, it is inefficient to directly use raw pixels for self-supervised pretraining. Specifically, remote sensing images typically contain a large number of pixels that are nearly identical. Involving all of these pixels in the training process may not enhance the network's performance, and instead, it may result in extra computational expenses. The superpixel theory, which has gained wide acceptance in classification and image segmentation tasks, can notably decrease computational expenses [34]. During the self-supervised pretraining stage, we use the simple linear iterative clustering (SLIC) algorithm [35] to oversegment the original pixels into n_p superpixels. In each superpixel, taking the pixel closest to the geometric center of as the target pixel,

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a 16 × 16 patch is constructed with its neighborhood as the input for the network. By using this method, the number of pretraining samples is reduced from hundreds of thousands pixels to n_p superpixels, without compromising the quality of the extracted features. In the experiments, the number of superpixels n_p is set as 20 000. However, using superpixels results in a sacrifice in spatial resolution. To better illustrate the performance of the proposed feature extraction network in pixelwise tasks, we use superpixels only during the pretraining stage. In the final supervised classification step, we still use the original individual pixels for training and testing.

Few-Shot Classification: To validate the performance of features extracted by the proposed TACL model, a series of few-shot land cover classification experiments with a simple classifier support vector machine (SVM) are carried out. In the classification stage, only five samples in each class are randomly selected as the training set. During the comparison of different methods, it is worth noting that they are all trained on the same training set. Furthermore, in consideration of the stochastic nature of the random selection, the experiments are repeated ten times, with a new set of randomly selected training samples for each run. The quantitative metrics to measure the classification performance are the overall accuracy (OA), the kappa coefficient (κ), the average accuracy (AA), and the mean of OAs of 10 runs (mOA).

Parameter Setting: The proposed self-supervised network is optimized by the ADAM optimizer, in which the learning rate is set as 10^{-4}. The number of epochs is set as 200, and the batch size is 512. Similar to MOCO and SimCLR, the temperature parameter τ is set as 0.07. The numbers of SCNB $[n_1, n_2, n_3]$ in the view-specific encoders are set as [4, 4, 4]. The original size of input patches $H \times W$ is set as 13 × 13, and then, they will be upsampled to a size of 16 × 16. A larger patch size implies more spatial contextual information, but at the same time, it also introduces more irrelevant redundancy. The influence of H and W values on OAs in the three datasets will be analyzed in Section IV-E.

Comparison With Existing Methods: Through self-supervised pretraining, the proposed method, with the simplest classifier SVM, can achieve comparable or even better performance than current state-of-the-art (SOTA) supervised classification networks. To verify this, we compare it with several advanced single-model and multimodal supervised classification models SpectralFormer [36], the multimodal transformer network (MTNet) [37], and the cross-channel reconstruction network (CCR-Net) [38]. Furthermore, we also incorporated several competitive multimodal self-supervised pretraining methods, including the self-supervised SAR-optical fusion (SSOF) [20], the CMCL model [24], the self-supervised feature representation (SSFL) model [39], the DECL model [25], the contrastive masked image autoencoder (CoMAE) [19], and the CCL model [26]. It should be noted that the network structures of some of the methods, such as SSOF and CoMAE, were not designed for small-sized inputs such as 16 × 16. Therefore, we make modifications to the input layers of these networks to match the size of the input data. In particular, for CoMAE, the processing object of the position embedding layers is no longer the patches in the original article, but the individual pixels. In addition, some of these self-supervised methods adopted a carefully designed classification network after pretraining to improve their performance. To ensure a fair comparison, we treat these self-supervised methods as feature extractors and directly input the extracted features into SVM for classification. For example, in the case of CCL, we use the CCL-FT model mentioned in [26], i.e., the version without fine-tuning network, for comparison.

C. Few-Shot Classification Results

The few-shot classification results obtained with different methods of the three datasets are shown in Figs. 8–10. Also, the quantitative evaluation of the results can be found in Tables II–IV.

As can be seen in Tables II–IV, due to the complementarity of multimodal data, multimodal methods achieve better performance than single-model methods. For example, both being transformer-based models, MTNet achieved a 4.71% increase in OA over SpectralFormer on the Houston 2013 dataset, a 1.85% increase in OA on the Trento dataset, and a 6.54% increase in OA on the Augsburg dataset. In addition, benefiting from the pretraining process on a large number of unlabeled samples, all the self-supervised methods can achieve high classification accuracy with only an SVM classifier, despite differences in their detailed designs. Even compared to the SOTA supervised networks, their performance is still competitive. Apart from the proposed TACL, the best-performing self-supervised method on the Houston 2013 dataset is CCL-FT with an OA of 83.61%, which is very close to the OA of the best-performing supervised method MTNet. Likewise, DECL on the Trento dataset and CoMAE on the Augsburg dataset also exhibit performance comparable to the best-performing supervised methods, with OAs of 96.93% and 74.92%, respectively.

The proposed TACL model, with the triplet contrastive learning framework and the AHSG strategy, achieves the highest OA, as well as other quantitative metrics. Specifically, TACL achieves an OA of 85.26% on the Houston 2013 dataset, which is 1.40% higher than the best-performing supervised model MTNet and 1.65% higher than the self-supervised model CCL-FT. Despite not attaining the highest accuracy on certain categories, TACL exhibits the utmost average accuracy across all categories, surpassing 70% accuracy across all categories. This distinguishes it from other approaches that may exhibit high accuracy in specific categories while displaying low accuracy in others. Similarly, on the Trento dataset, TACL achieves an OA of 97.57%, which is 0.49% higher than the best-performing supervised model MTNet and 0.64% higher than the self-supervised model DECL. On the Augsburg dataset, TACL also reaches the highest OA of 77.35%.

Few-shot classification tasks heavily rely on the quality of training samples, and the results on a particular small training set may not guarantee the method's transferability. Therefore, we conducted experiments on ten randomly selected training sets and evaluated the models' performance using the average OA of the ten experiments, i.e., mOA. On the three datasets,

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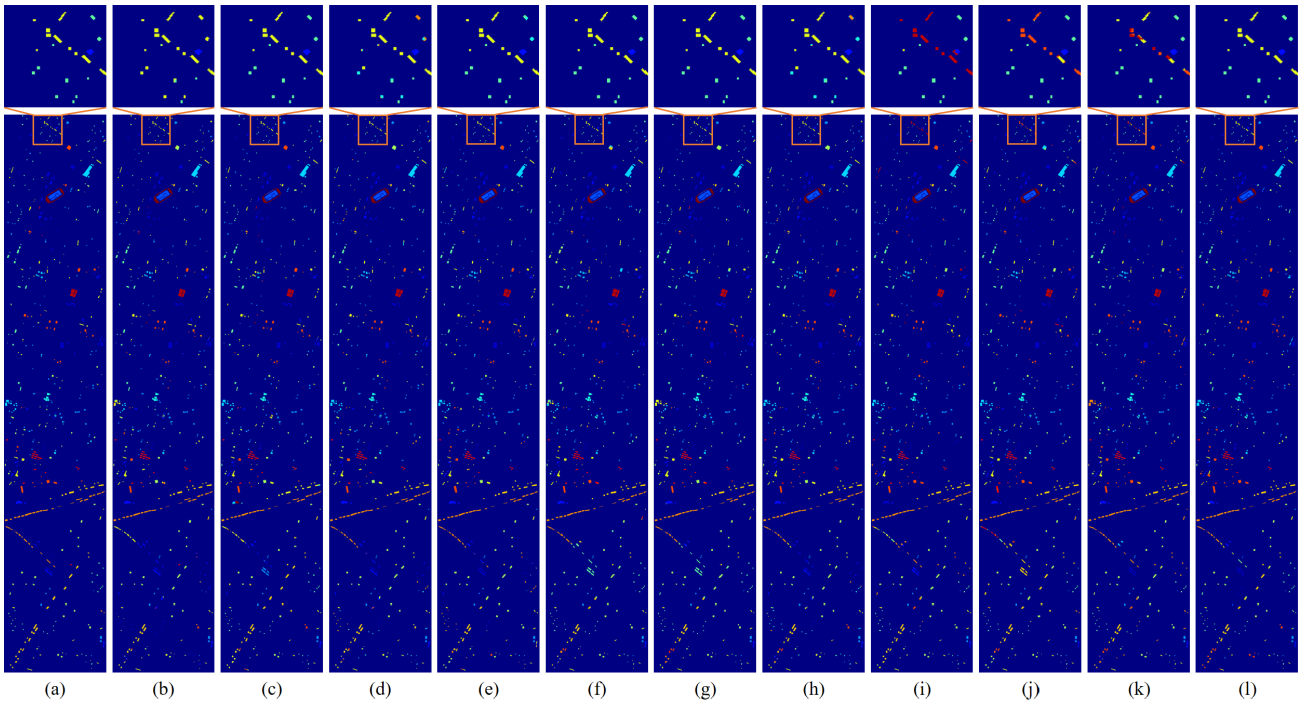


Fig. 8. Few-shot classification results of different methods of the Houston 2013 HSI-LiDAR dataset. (a) Ground truth. (b) SVM. (c) SpectralFormer. (d) CCR-Net. (e) MTNet. (f) SSOF. (g) CMCL. (h) SSFL. (i) DECL. (j) CoMAE. (k) CCL-FT. (l) TACL.

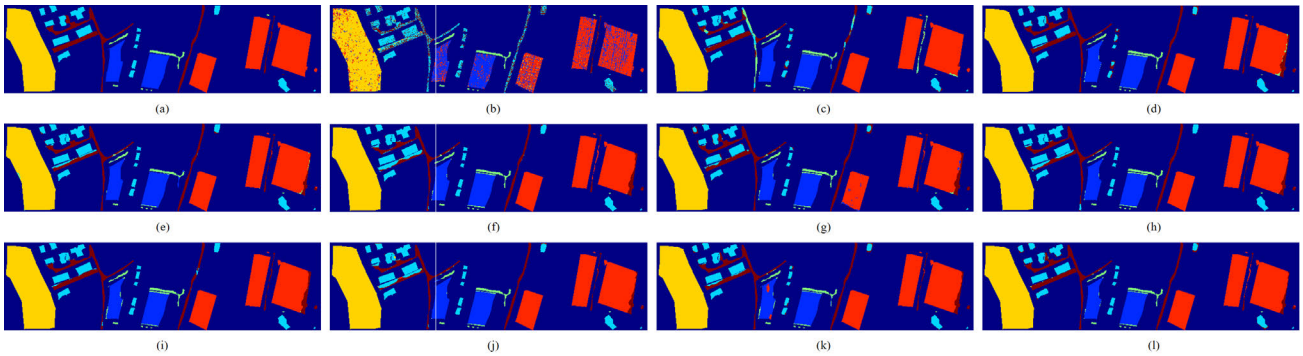


Fig. 9. Few-shot classification results of different methods of the Trento HSI-LiDAR dataset. (a) Ground truth. (b) SVM. (c) SpectralFormer. (d) CCR-Net. (e) MTNet. (f) SSOF. (g) CMCL. (h) SSFL. (i) DECL. (j) CoMAE. (k) CCL-FT. (l) TACL.

TACL has the highest mOAs among all methods, which are 84.29%, 97.39%, and 77.35%.

On the Augsburg dataset, the mOA metric also reveals the limitations of supervised methods in few-shot classification. Although MTNet can achieve higher OA on specific random training sets compared to most self-supervised + SVM methods, its mOA is significantly lower than that of all self-supervised methods. This indicates that the direct use of supervised methods for classification is profoundly affected by the quality of the training data, leading to considerable performance disparities across diverse training sets. Conversely, self-supervised pretraining adeptly mitigates this challenge, bolstering the classifier's robustness. This reinforces the notion that self-supervised pretraining is better suited for few-shot tasks.

Overall, the experimental results demonstrate the superiority and robustness of TACL in few-shot classification tasks, indicating its potential in various real-world applications.

D. Ablation Study

Triple Contrastive Learning Framework: In the context of contrastive learning for two modalities of remote sensing images, three distinct approaches may be envisioned. The first approach involves treating each modality as an independent view, followed by direct contrastive learning without view fusion, which is referred to as double-view contrastive learning (DCL) here. The second approach involves fusing the two views and generating two augmented views from the fused representation, and here, it is referred to as fused-view contrastive learning (FCL). The third approach, which is adopted in this article, is the triple-view contrastive learning (TCL), and it involves the contrastive learning of the fused view with the two original views separately. Table V provides a comparative performance analysis of the three approaches on three datasets. Also, the results of single-modal contrastive learning (SCL), which is referred to as transforming single-modal HSIs into two views via data augmentation, are also listed in the table

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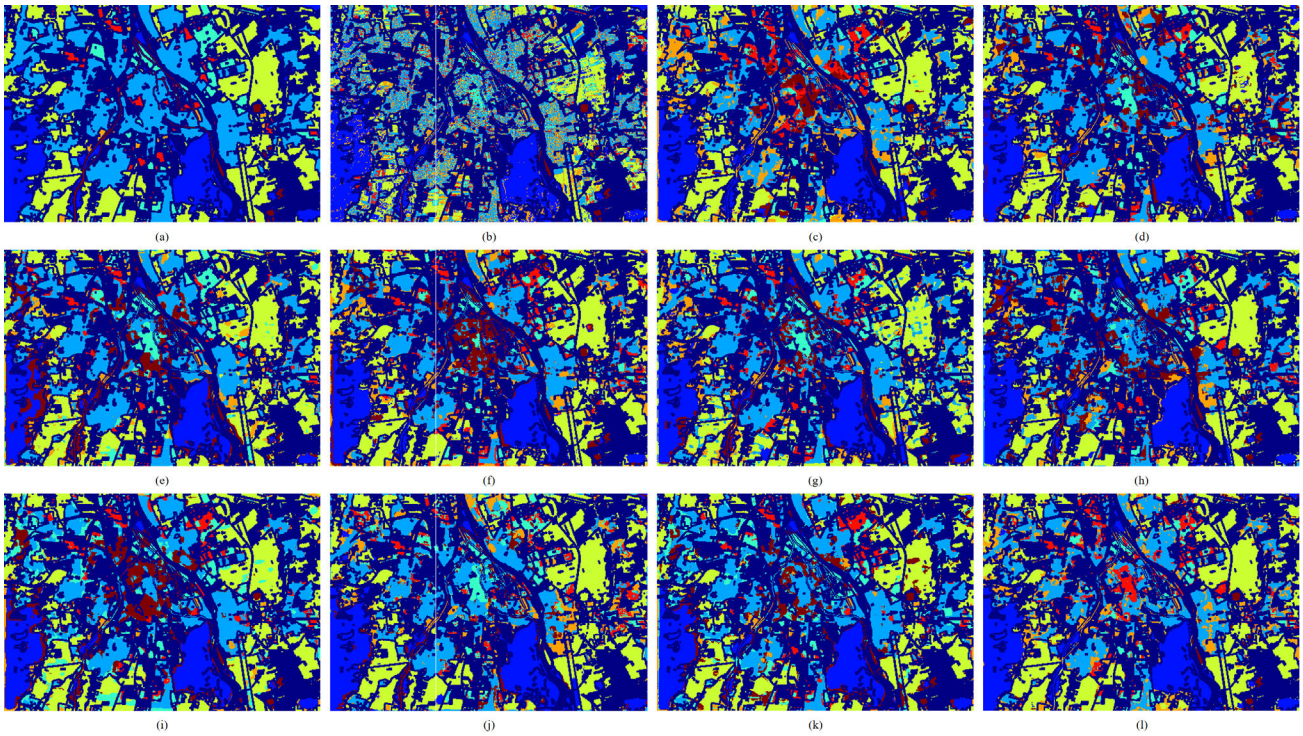


Fig. 10. Few-shot classification results of different methods of the Augsburg HSI-SAR dataset. (a) Ground truth. (b) SVM. (c) SpectralFormer. (d) CCR-Net. (e) MTNet. (f) SSOF. (g) CMCL. (h) SSFL. (i) DECL. (j) CoMAE. (k) CCL-FT. (l) TACL.

TABLE II

QUANTITATIVE EVALUATION OF THE DIFFERENT METHODS OF THE HOUSTON 2013 HSI-LIDAR DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Class	Supervised Classification				Self-supervised Pre-training + SVM						
		SVM	Spectral Former	CCR-Net	MTNet	SSOF	CMCL	SSFL	DECL	CoMAE	CCL-FT	TACL
Accuracy (%)	1	91.53	78.90	92.89	91.69	77.14	69.38	**93.69**	85.13	84.97	90.33	83.37
	2	76.32	64.43	87.88	74.64	84.61	84.29	**88.12**	77.59	74.72	73.68	75.60
	3	99.43	100.0	99.43	99.86	100.00	100.0	99.71	99.86	98.85	99.71	**100.0**
	4	97.83	98.71	97.59	98.47	97.75	97.91	97.35	97.59	**100.0**	97.83	91.56
	5	84.38	87.68	99.92	**100.0**	90.26	80.19	99.92	98.79	99.03	79.47	99.28
	6	82.77	77.54	**87.69**	85.54	85.85	85.85	90.46	85.85	85.85	85.85	85.85
	7	51.03	97.56	49.37	69.72	83.36	**99.05**	49.05	92.51	94.72	91.80	98.11
	8	81.67	70.98	88.42	86.33	58.36	**89.07**	85.45	66.48	80.79	82.88	71.06
	9	84.90	83.71	80.03	77.88	86.74	**92.09**	82.99	66.13	64.54	65.02	73.80
	10	57.05	50.37	57.05	61.94	72.53	66.91	57.05	71.39	71.80	57.05	**78.48**
	11	69.31	69.07	89.88	92.63	87.29	85.75	86.72	80.32	71.98	**100.0**	86.40
	12	49.72	63.58	48.66	64.72	52.07	37.88	51.09	67.72	54.18	76.89	**80.45**
	13	87.42	84.43	83.37	79.74	90.62	**94.67**	83.16	83.37	89.13	83.80	86.35
	14	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.00	100.0	**100.0**
	15	96.06	88.94	96.52	97.27	88.48	86.52	97.73	93.03	**99.70**	94.09	86.06
OA (%)		77.82	79.15	81.78	83.86	81.54	82.65	81.88	82.63	82.49	83.61	**85.26**
AA (%)		80.63	81.06	83.91	85.36	83.67	84.64	84.17	84.38	84.68	85.23	**86.42**
κ		0.760	0.775	0.803	0.826	0.801	0.813	0.805	0.812	0.811	0.823	**0.841**
mOA (%)		76.91	78.59	79.69	82.33	80.62	79.95	81.01	81.50	81.63	82.56	**84.29**

for comparison. It should be noted that all of them share a common encoder architecture and employ the AHSG strategy. Moreover, the view fusion module of FCL is identical to that of TCL. It can be observed that the TCL approach achieves the best performance on both datasets, demonstrating its superiority. In addition, both the TCL and FCL approaches, which integrate view fusion, outperform the nonfusion or single-modal methods. This validates the effectiveness of view fusion as a viable approach to avoid contrastive semantic conflicts between views.

AHSG Strategy: In Fig. 11, we use the t-distributed stochastic neighbor embedding (t-SNE) [40], a widely used nonlinear dimensionality reduction technique for data visualization, to visualize the features produced by contrastive learning. It can help observe the differences between the extracted features with and without the AHSG strategy. In the figure, different colors are assigned to all points according to their GT labels. It can be seen that in the t-SNE map of the original HSI, the points of different classes are mixed together and difficult to be separated. However, contrastive learning succeeds in

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TABLE III

QUANTITATIVE EVALUATION OF THE DIFFERENT METHODS OF THE TRENTO HSI-LIDAR DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Class	Supervised Classification				Self-supervised Pre-training + SVM						
		SVM	Spectral Former	CCR-Net	MTNet	SSOF	CMCL	SSFL	DECL	CoMAE	CCL-FT	TACL
Accuracy (%)	1	84.04	97.32	96.65	**98.74**	96.78	96.58	96.58	94.94	97.55	93.51	96.78
	2	84.29	93.87	91.25	98.59	99.24	91.08	99.62	**99.66**	98.52	95.28	93.39
	3	**98.75**	90.61	97.91	86.64	97.70	97.91	97.49	97.91	93.53	93.95	97.7
	4	89.56	99.81	100.0	99.62	100.0	99.99	99.99	99.92	100.0	100.0	**100.0**
	5	78.98	**99.51**	95.26	97.00	95.76	94.12	96.92	95.08	96.71	96.11	97.34
	6	44.52	67.14	96.16	88.19	75.90	95.02	83.77	94.30	81.25	93.86	**96.16**
OA (%)		80.06	95.23	96.63	97.08	95.46	96.08	96.69	96.93	96.32	96.59	**97.57**
AA (%)		80.02	91.38	96.21	94.79	94.23	95.78	95.73	96.97	94.59	95.45	**96.89**
κ		0.738	0.937	0.955	0.961	0.940	0.948	0.956	0.959	0.951	0.955	**0.968**
mOA (%)		78.18	94.65	96.10	96.92	95.02	95.84	96.50	96.78	96.05	96.23	**97.39**

TABLE IV

QUANTITATIVE EVALUATION OF THE DIFFERENT METHODS OF THE AUGSBURG HSI-SAR DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Methods	Class	Supervised Classification				Self-supervised Pre-training + SVM						
		SVM	Spectral Former	CCR-Net	MTNet	SSOF	CMCL	SSFL	DECL	CoMAE	CCL-FT	TACL
Accuracy (%)	1	94.84	90.69	88.95	74.21	84.54	83.50	87.75	78.63	92.34	81.82	**95.12**
	2	58.51	56.20	62.17	**72.38**	61.93	71.42	60.78	62.84	59.04	69.80	67.25
	3	37.76	35.68	58.61	**73.85**	47.49	59.28	55.73	59.70	61.08	59.34	46.66
	4	44.64	79.31	84.25	82.13	82.89	72.09	84.38	85.69	**90.12**	82.22	87.30
	5	81.04	82.26	83.65	79.13	84.87	69.39	69.74	56.00	77.22	57.74	**85.91**
	6	37.75	47.11	40.06	45.05	48.69	50.82	56.72	44.98	49.73	50.76	**62.13**
	7	43.53	42.03	44.05	59.28	55.69	51.37	56.73	**61.37**	30.39	48.04	36.54
OA (%)		58.43	68.79	73.53	75.33	72.08	72.30	73.18	72.79	74.92	74.71	**77.35**
AA (%)		56.87	61.90	65.96	**69.43**	66.59	65.41	67.40	64.17	65.70	64.25	68.70
κ		0.466	0.597	0.655	0.675	0.638	0.631	0.646	0.646	0.670	0.667	**0.701**
mOA (%)		45.56	55.42	65.71	66.58	68.56	71.42	71.11	68.87	71.8	71.42	**73.44**

TABLE V

ABLATION STUDY OF THE TRIPLE CONTRASTIVE LEARNING FRAMEWORK

Methods	Houston 2013		Trento		Augsburg	
	OA (%)	mOA (%)	OA (%)	mOA (%)	OA (%)	mOA (%)
SCL	81.05	80.56	96.02	95.84	71.95	68.17
DCL	83.32	82.21	96.95	96.78	74.28	70.66
FCL	84.88	83.81	97.10	96.98	75.13	71.48
TCL	**85.26**	**84.29**	**97.57**	**97.39**	**77.35**	**73.44**

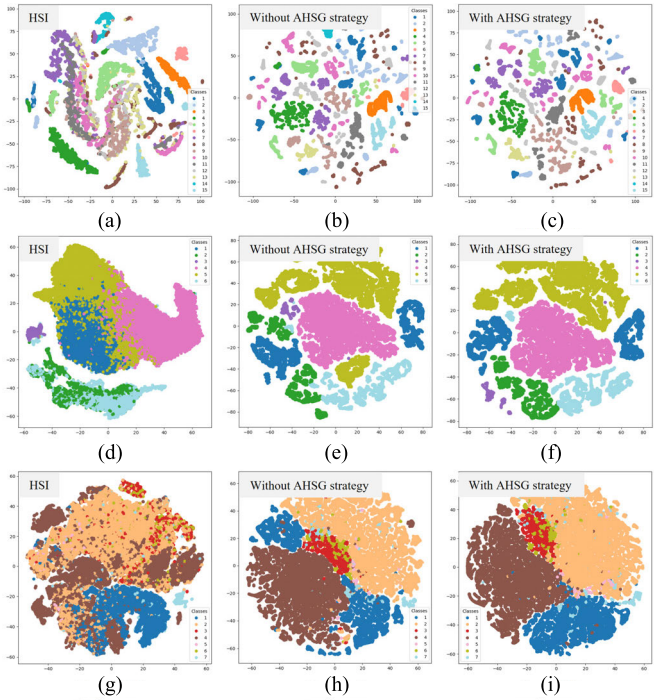


Fig. 11. Data visualization of extracted features using t-SNE without and with the AHSG strategy. (a)–(c) On the Houston 2013 dataset. (d)–(f) On the Trento dataset. (g)–(i) On the Augsburg dataset.

increasing the distances between different classes, making them more separable for classification tasks. To objectively evaluate the quality of t-SNE maps, we introduce a metric n_d, which measures, on average, the number of points that belong to a different class among the top 1% of each point's nearest neighbors. A larger n_d indicates that there are more samples of different classes present around each point. It is evident that for self-supervised feature extraction, a smaller n_d is desired. Here, because of the abundance of data points in the Augsburg dataset and taking computational memory into account, the t-SNE map incorporates only a randomly selected 1/2 of the data points. As a result of the AHSG process, the n_d value on the Houston 2013 dataset decreases from 22.72 to 20.71; on the Trento dataset, it decreases from 9.77 to 7.50; and on the Augsburg dataset, it decreases from 53.48 to 44.59. This observation confirms that the proposed AHSG strategy is effective in extracting features with improved separability.

Fig. 12 illustrates the impact of varying values of c_0 in the AHSG process on the classification performance. The parameter c_0 essentially represents the degree of preservation of the original distribution of negative feature vectors. Specifically, when c_0 is set as 1, it corresponds to the complete retention

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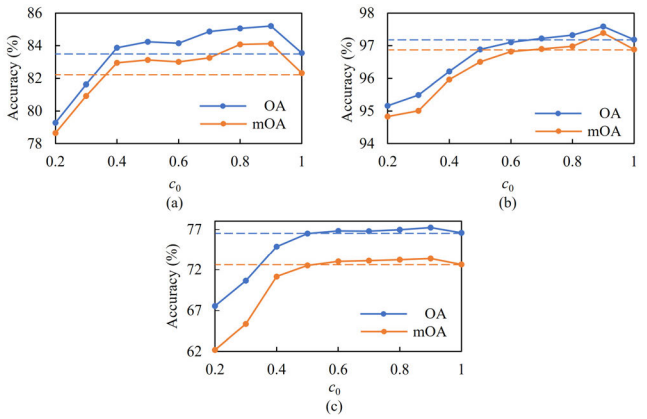


Fig. 12. Ablation study of the adversarial hard-negative sample generation strategy with varying values of the parameter c_0. (a) On the Houston 2013 dataset. (b) On the Trento dataset.

TABLE VI
ABLATION STUDY OF THE NETWORK STRUCTURE

Methods	Houston 2013		Trento		Augsburg	
	OA (%)	mOA (%)	OA (%)	mOA (%)	OA (%)	mOA (%)
TACL	**85.26**	**84.29**	**97.57**	**97.39**	**77.35**	**73.44**
w/o CVAB	85.03	83.98	97.19	97.05	77.11	73.26
w/o CFEB	84.56	83.48	96.53	96.42	76.17	72.39
w/o SCNB	85.15	84.11	97.32	97.16	77.31	73.36

of the original distribution without the AHSG process. Taking the accuracy at $c_0 = 1$ as the baseline, it can be observed that for the Houston 2013 dataset, both OA and mOA are higher for $c_0 > 0.4$ compared to the baseline. When c_0 is too small, the feature vectors of different negative samples become excessively similar, leading to an overly concentrated data distribution. Consequently, this ultimately results in a significant decline in both OA and mOA. Similar phenomena hold true for the Trento dataset and the Augsburg dataset, whereby the accuracies surpass the baseline when $c_0 > 0.6$, while falling below the baseline when $c_0 < 0.6$. This observation substantiates the effectiveness of the proposed AHSG strategy in enhancing the performance of contrastive learning.

Design of Network Structure: Table VI illustrates the performance of the proposed TACL model under different configurations, including without the CVAB, the CFEB, and the SCNB. The configuration "w/o SCNB" refers to replacing the SCNB with the original CNB, which preserves the 7×7 large-scale convolution instead of the 3×3 convolution in SCNB. It can be observed that under these three configurations, both OA and mOA of the three datasets exhibit varying degrees of decline. Without CVAB, achieving high accuracy is still possible by simply concatenating the features of the two views as the fused view. Obviously, the spatial information coupling introduced by CVAB can further enhance the performance of the network. The removal of CFEB, which results in the largest decrease, highlights the significance of focusing on central features in the encoders. The adoption of SCNB, compared to CNB, resulted in almost identical OA, indicating that for 16×16 input patches, larger convolution kernels are indeed not necessarily essential in our network.

TABLE VII
CLASSIFICATION PERFORMANCE WITH VARIOUS INPUT PATCH SIZES

Patch size	Houston 2013		Trento		Augsburg	
	OA (%)	mOA (%)	OA (%)	mOA (%)	OA (%)	mOA (%)
7	81.16	79.95	95.32	95.67	72.33	68.26
9	83.44	82.30	96.55	96.29	75.21	71.38
11	85.21	84.20	97.42	97.31	76.94	73.06
13	**85.26**	**84.29**	**97.57**	**97.39**	**77.35**	**73.44**
15	84.52	83.22	97.15	96.93	77.13	73.20

TABLE VIII
CLASSIFICATION PERFORMANCE WITH VARIOUS NUMBERS OF SLIGHT CONVNEXT BLOCKS (SCNB)

$[n_1, n_2, n_3]$	Houston 2013		Trento		Augsburg		# Paras. (M)	Flops (G)
	OA (%)	mOA (%)	OA (%)	mOA (%)	OA (%)	mOA (%)		
[2, 2, 2]	85.15	84.22	97.46	97.21	77.22	73.32	**6.936**	**138.8**
[4, 4, 4]	**85.26**	**84.29**	97.57	97.39	**77.35**	**73.44**	11.10	247.4
[8, 4, 4]	85.10	84.13	97.49	97.27	77.18	73.25	11.51	320.3
[4, 8, 4]	85.14	84.11	97.51	97.34	77.27	73.33	12.70	319.7
[4, 4, 8]	85.22	84.18	**97.59**	**97.40**	77.31	73.39	17.44	319.3
[8, 8, 8]	85.02	84.07	97.31	97.15	77.06	73.16	19.44	464.6

TABLE IX
CLASSIFICATION PERFORMANCE WITH VARIOUS NUMBERS OF NEGATIVE SAMPLES

# Negative samples	Houston 2013		Trento		Augsburg	
	OA (%)	mOA (%)	OA (%)	mOA (%)	OA (%)	mOA (%)
100	84.52	83.43	97.11	96.92	76.17	72.21
200	85.07	84.09	97.50	97.31	77.07	73.10
300	**85.26**	**84.29**	97.57	97.39	**77.35**	**73.44**
400	85.17	84.14	**97.61**	**97.42**	77.18	73.31
500	84.95	83.92	97.26	97.05	76.96	72.83

E. Parameter Analysis

Patch Size: Table VII illustrates the impact of patch size on classification accuracy, with the caveat that all of them will be upsampled to 16×16 before being fed into the network. Both datasets exhibit optimal performance in the range of 11–15, with the highest OA and mOA achieved when the patch size is 13. Therefore, we set the patch size as 13 for all our experiments.

Design of Encoders: The accuracy under varying numbers of SCNBs at each stage $[n_1, n_2, n_3]$ is presented in Table VIII. For reference, the renowned ResNet-50 model is composed of about 23.5M parameters, which indicates the remarkably lightweight profile of the proposed encoder structure. Taking the combination [4, 4, 4] as the baseline, it can be observed that increasing the number of blocks does not yield a significant improvement in accuracy. Conversely, it gives rise to a linear increment in the number of parameters and flops. Therefore, the combination of [4, 4, 4] is adopted for the design of encoders.

Number of Negative Samples: Table IX reports the results of varying the number of negative samples, in which all the negative samples undergo the AHSG process with $c_0 = 0.95$. In general, if the computational cost is not considered, increasing the number of negative samples for contrastive learning

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would enhance the performance. However, this table reveals an anomalous phenomenon, where the optimal performance is achieved at 300 and 400 samples for the three datasets, and further increasing the sample number to 500 leads to a slight degradation. This implies that the spatial information of remote sensing images is highly correlated, and a subset of representative pixels suffices to serve as negative samples. Further increasing the sample number does not enhance the sample diversity but rather incurs higher computational cost, which is detrimental to contrastive learning. Based on these considerations, we set the number of negative samples as 300.

V. Conclusion

In this article, the TACL model for self-supervised feature extraction of multimodal remote sensing images is proposed. Considering the contrastive semantic conflicts across different views, TACL aligns two original views and a fused view to construct the triplet contrastive learning framework. Through the AHSG strategy, TACL transforms the negative samples to be more challenging to discriminate from anchors with the same network. Furthermore, a central feature enhanced encoder with the SCNB is designed as the backbone of the TACL model. A series of experiments are conducted to validate the applicability of the features extracted by TACL on few-shot multimodal remote sensing image classification tasks.

The proposed method has significant implications for remote sensing applications, facilitating more accurate and efficient analysis in various domains. However, it still has room for further improvement. For example, the view fusion module as well as the encoders can be designed in a more sophisticated manner, and the sharing of parameters among different views can be employed for specific modules to alleviate computational burden, which will be in our future work.

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